Exercise 2

```
bimdika@Asus:~/lab8$ cd ~/lab8
bimdika@Asus:~/lab8$ ls ~/lab8
ex2_sum  ex2_sum.cc
bimdika@Asus:~/lab8$ mpirun -np 1 ./ex2_sum
Processes: 1  Sum: 50000005000000  Time: 0.024339 s
bimdika@Asus:~/lab8$ mpirun -np 2 ./ex2_sum
Processes: 2  Sum: 50000005000000  Time: 0.020700 s
bimdika@Asus:~/lab8$ mpirun -np 4 ./ex2_sum
Processes: 4  Sum: 50000005000000  Time: 0.007202 s
bimdika@Asus:~/lab8$ mpirun -np 8 ./ex2_sum
```

```
bimdika@Asus:~/lab8$ mpirun --oversubscribe -np 8 ./ex2_sum
Processes: 8  Sum: 50000005000000  Time: 0.007516 s
bimdika@Asus:~/lab8$ for p in 1 2 4 8; do for r in 1 2 3 4 5; do mpirun -np $p ./ex2_sum; done; done | awk '{s[$2]+=$6;
n[$2]++} END {for (p in s) printf "%s processes: avg %.6f s\n", p, s[p]/n[p]}' | sort -n
```

```
1 processes: avg 0.025586 s
2 processes: avg 0.013761 s
4 processes: avg 0.006991 s
bimdika@Asus:~/lab8$
```

Exercise 3

```
bimdika@Asus:~/SE3082_Lab08_IT24103435$ cat > ~/SE3082_Lab08_IT24103435/exercise3/ex3_pi.cc << 'EOF'
// ex3_pi.cc - Monte Carlo estimate of Pi with MPI
#include <mpi.h>
#include <cstdio>
```

```
bimdika@Asus:~/SE3082_Lab08_IT24103435$ mpicxx ~/SE3082_Lab08_IT24103435/exercise3/ex3_pi.cc -o ~/SE3082_Lab08_IT2410343
5/exercise3/ex3_pi
bimdika@Asus:~/SE3082_Lab08_IT24103435$ mpirun -np 1 ~/SE3082_Lab08_IT24103435/exercise3/ex3_pi
Processes: 1  Pi: 3.141386  Time: 0.174225 s
bimdika@Asus:~/SE3082_Lab08_IT24103435$ mpirun -np 2 ~/SE3082_Lab08_IT24103435/exercise3/ex3_pi
Processes: 2  Pi: 3.141265  Time: 0.106119 s
bimdika@Asus:~/SE3082_Lab08_IT24103435$ mpirun -np 4 ~/SE3082_Lab08_IT24103435/exercise3/ex3_pi
Processes: 4  Pi: 3.140876  Time: 0.045852 s
bimdika@Asus:~/SE3082_Lab08_IT24103435$ mpirun --use-hwthread-cpus -np 8 ~/SE3082_Lab08_IT24103435/exercise3/ex3_pi
Processes: 8  Pi: 3.141124  Time: 0.030410 s
bimdika@Asus:~/SE3082_Lab08_IT24103435$
```